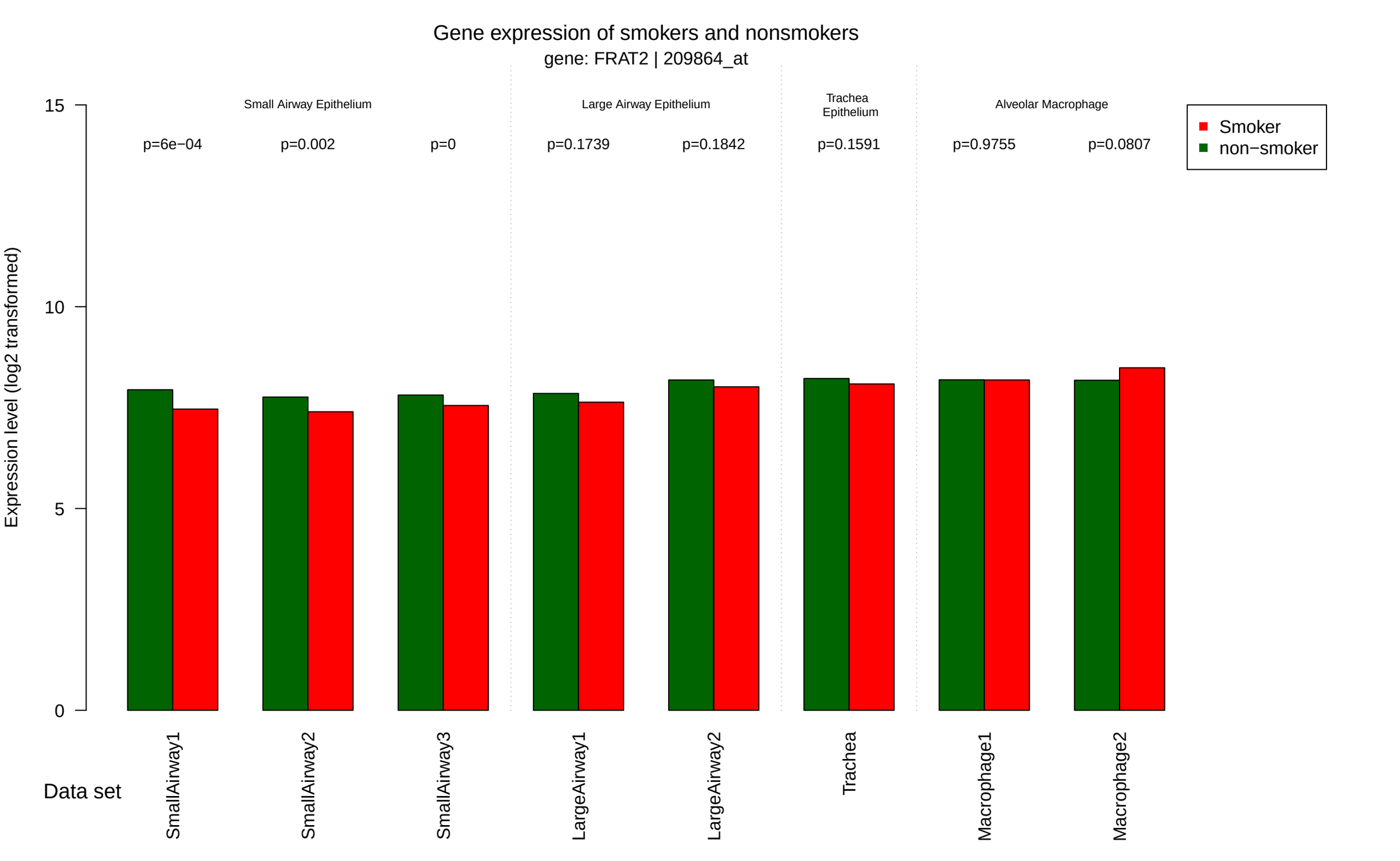
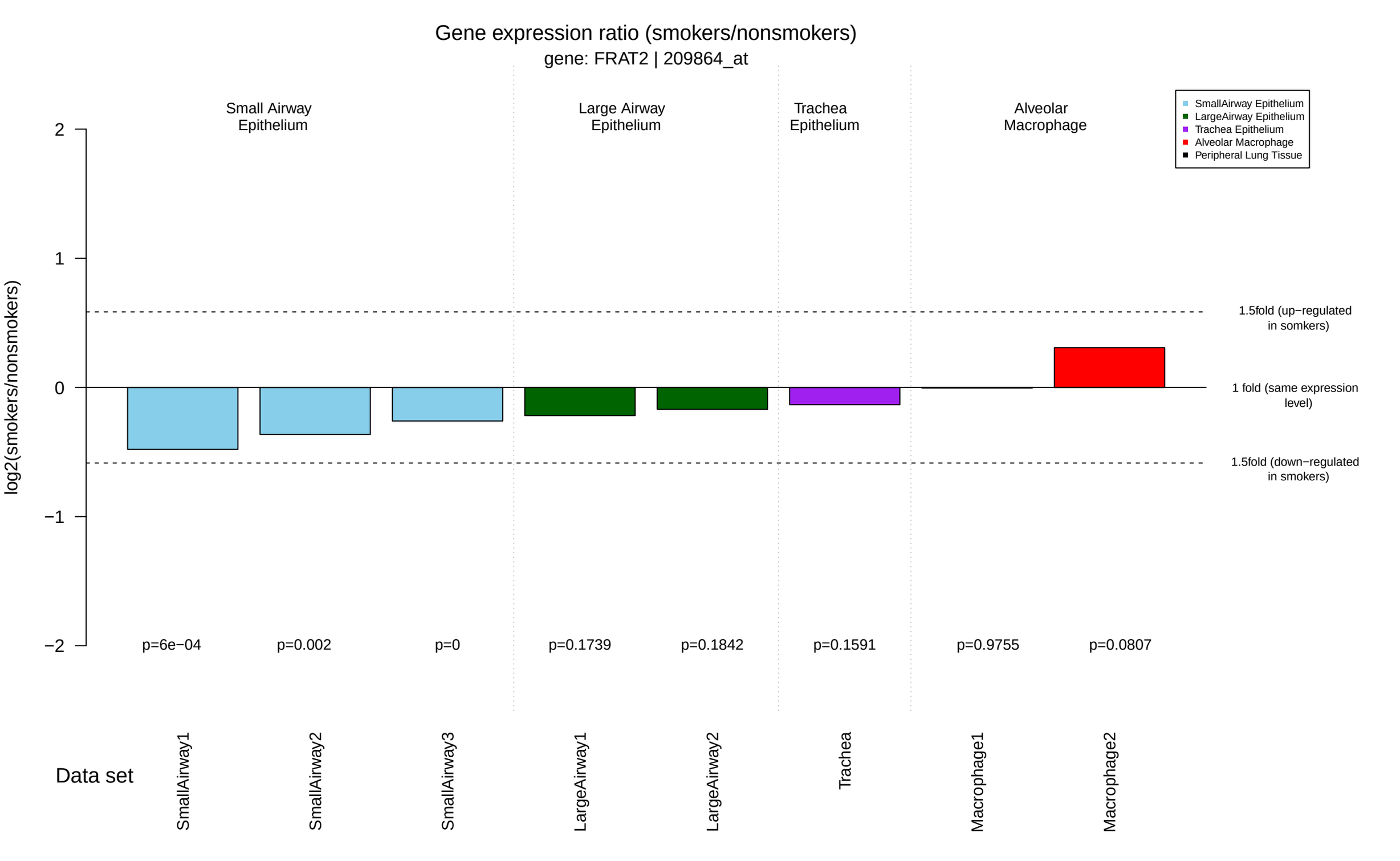
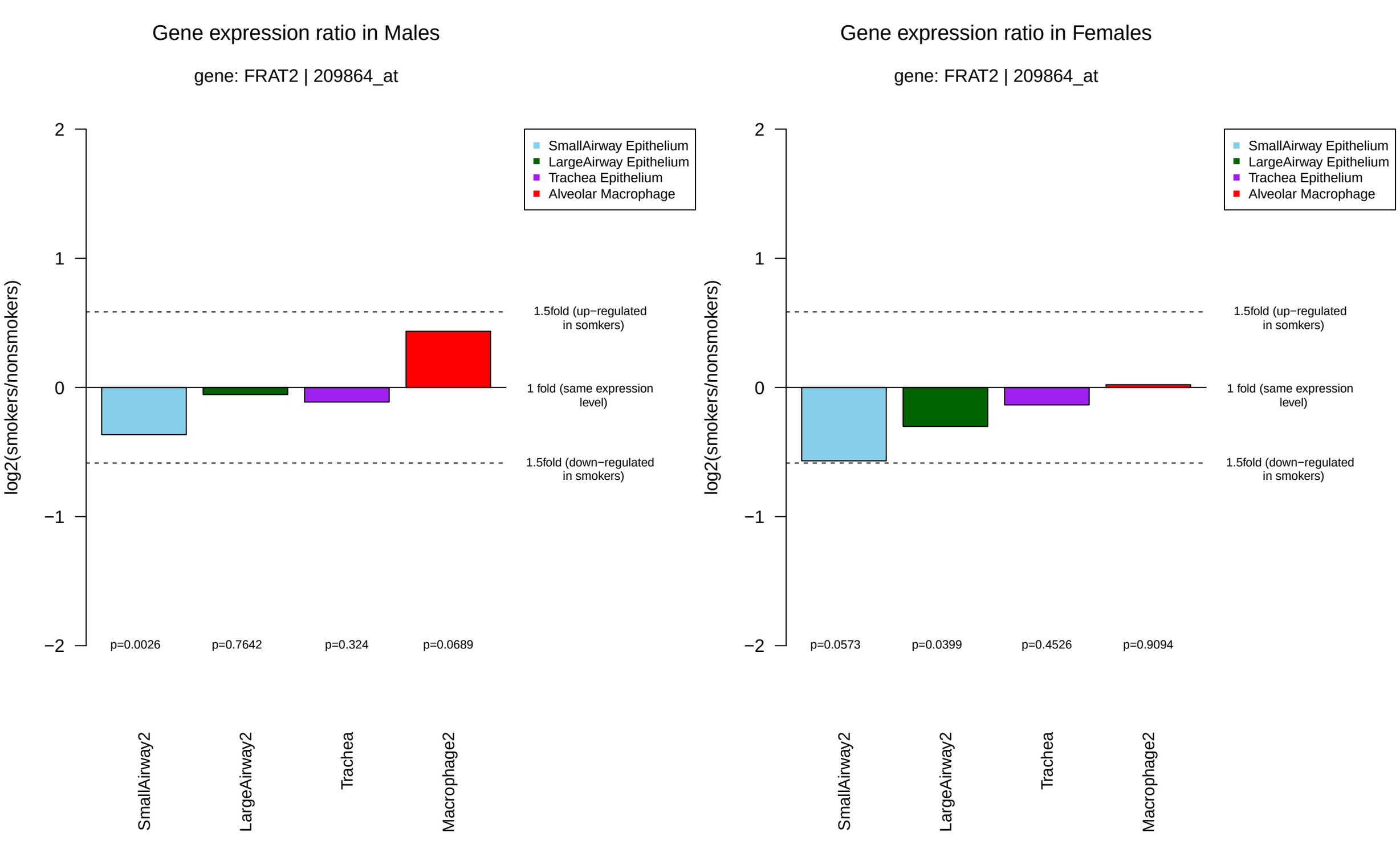
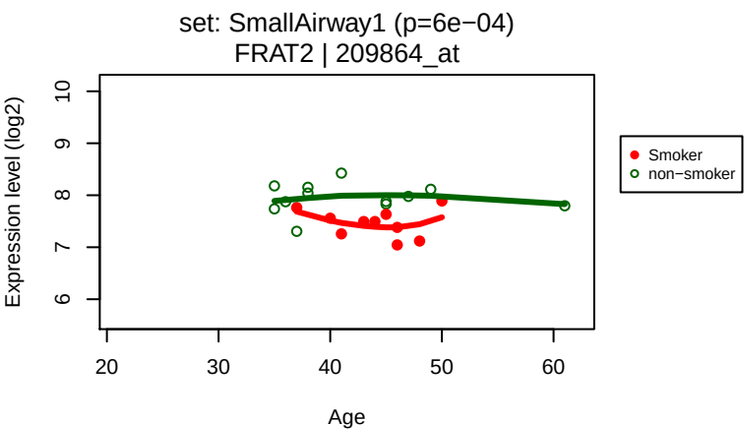
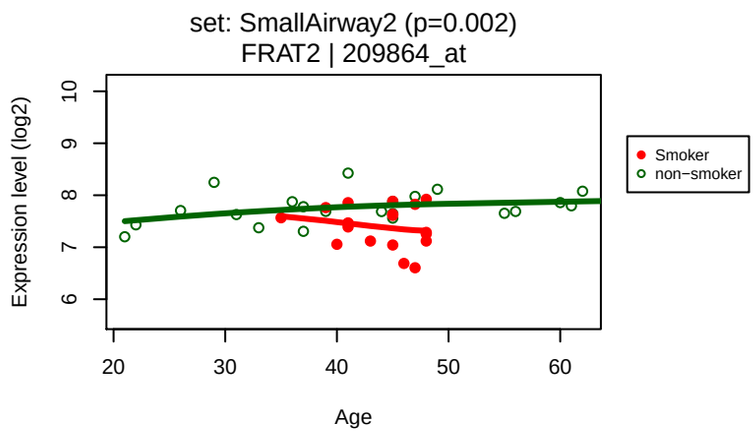
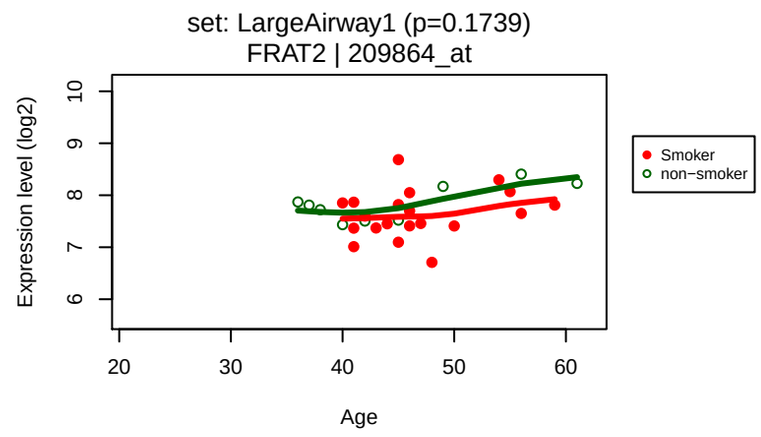

Gene expression of smokers and nonsmokers
gene: FRAT2 | 209864_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=6e−04
p=0.002
p=0
p=0.1739
p=0.1842
p=0.1591
p=0.9755
p=0.0807
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

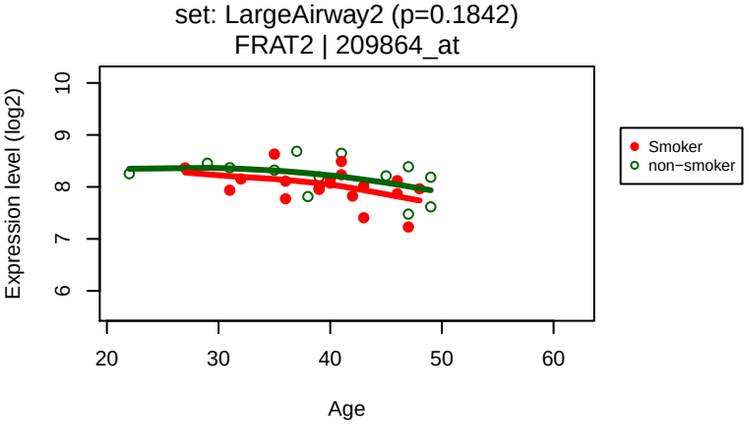

Gene expression ratio (smokers/nonsmokers)
gene: FRAT2 | 209864_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=6e−04
p=0.002
p=0
p=0.1739
p=0.1842
p=0.1591
p=0.9755
p=0.0807
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

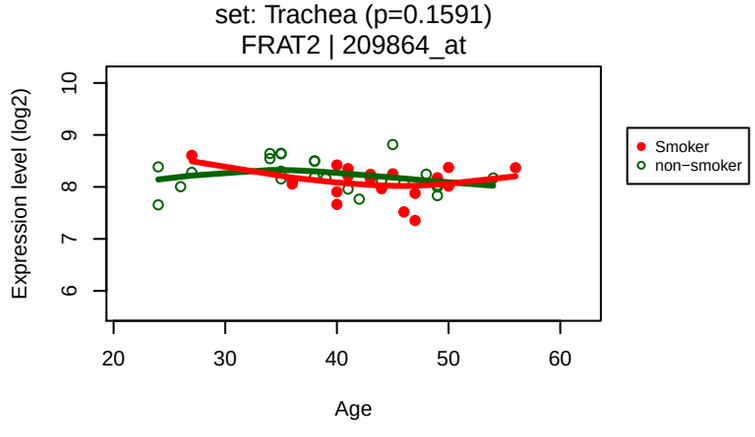

Gene expression ratio in Males
gene: FRAT2 | 209864_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0026
p=0.7642
p=0.324
p=0.0689
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: FRAT2 | 209864_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0573
p=0.0399
p=0.4526
p=0.9094
SmallAirway2
LargeAirway2
Trachea
Macrophage2

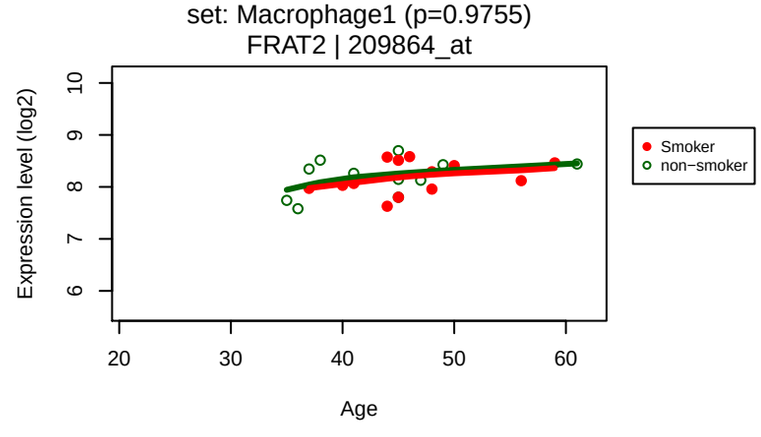

set: SmallAirway1 (p=6e−04)
FRAT2 | 209864_at
Expression level (log2)
Age
Smoker
non−smoker

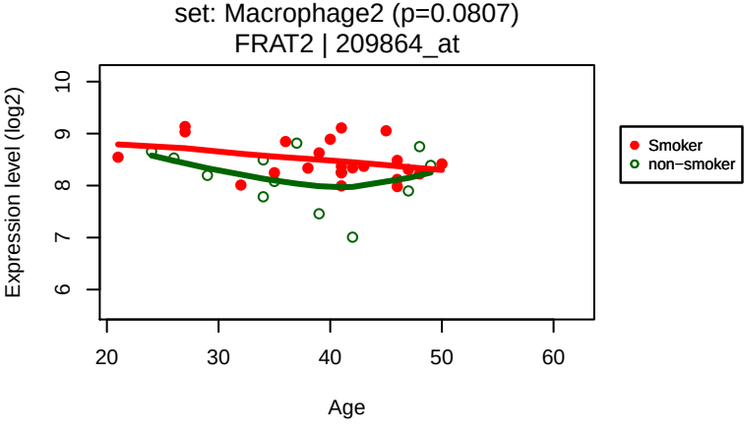

set: SmallAirway2 (p=0.002)
FRAT2 | 209864_at
Expression level (log2)
Age
Smoker
non−smoker

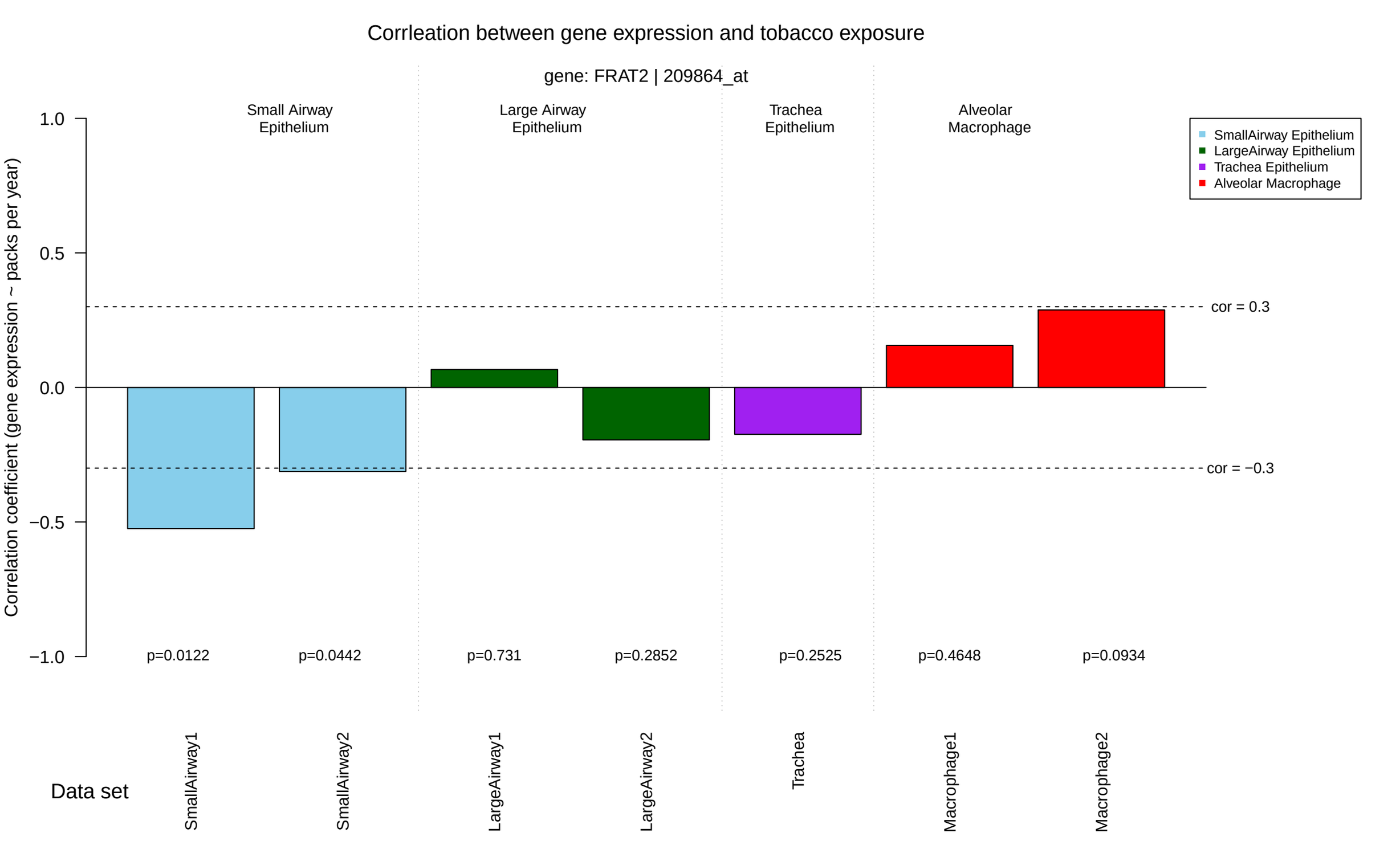

set: LargeAirway1 (p=0.1739)
FRAT2 | 209864_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1842)
FRAT2 | 209864_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1591)
FRAT2 | 209864_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9755)
FRAT2 | 209864_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0807)
FRAT2 | 209864_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: FRAT2 | 209864_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0122
p=0.0442
p=0.731
p=0.2852
p=0.2525
p=0.4648
p=0.0934
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2